

Project Design Phase-II

Data Flow Diagram & User Stories

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Data Flow Diagram

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level-0 (Context Diagram)

User (Browser) → [HTTPS/JSON] → React + Vite Client → [Axios REST calls] → Express.js REST API (JWT auth middleware → Zod validation → Controllers) → [Mongoose ODM] → MongoDB (Users, Foods, Recipes, Meals, NutritionLogs, WaterLogs, Favorites, Settings). The API layer also calls two external services: the Google Gemini API for AI chat responses, and Cloudinary for image storage (with local disk as a fallback).

User Stories

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Priority	Release
End User	Registration	USN-1	As a user, I can register for the application by entering my name, email, password and confirming my password.	High	Sprint-1
End User	Login	USN-2	As a user, I can log into the application by entering my email & password and receive a JWT access & refresh token.	High	Sprint-1
End User	Password Recovery	USN-3	As a user, I can request a password reset token and set a new password if I forget mine.	Medium	Sprint-1
End User	Profile Management	USN-4	As a user, I can view and update my profile (age, gender, height, weight, goal, activity level, diet preference, allergies) and upload a profile photo.	High	Sprint-1
End User	BMI Calculator	USN-5	As a user, I can calculate my BMI using metric or imperial units and see category-based guidelines.	High	Sprint-2
End User	Calorie Calculator	USN-6	As a user, I can calculate my BMR/TDEE based on my activity level and apply a calculated calorie goal directly to my profile.	High	Sprint-2
End User	Food Tracking	USN-7	As a user, I can log food consumed under a meal slot (breakfast/lunch/dinner/snack) and see my daily macro totals update automatically.	High	Sprint-2
End User	Water Tracking	USN-8	As a user, I can log water intake entries and track my progress toward my daily water goal.	High	Sprint-2
End User	Recipes	USN-9	As a user, I can search, filter and paginate recipes, and mark recipes as favorites.	Medium	Sprint-3
End User	Meal Planning	USN-10	As a user, I can plan meals ahead of time using the Meal Planner.	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story No.	User Story / Task	Priority	Release
End User	AI Assistant	USN-11	As a user, I can chat with the AI Nutrition Assistant to get personalized dietary guidance.	High	Sprint-3
Administrator	Admin Panel	USN-12	As an admin, I can view platform statistics, manage users, and curate standard system foods and recipes.	Medium	Sprint-4